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Clinical study

Standard awake surgery versus hypnosis aided awake surgery for the management of high grade gliomas: A non-randomized cohort comparison controlled trial

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ABSTRACT

Hypnosis could extend the time of Intraoperative Neuropsychological Testing and Brain Mapping in Awake Surgery. A clinical validation for the Hypnosis aided AS (HAs) is still ongoing and further evidences are required. The objective of the present study is to compare two homogeneous cohorts of patients undergoing AS, the first with the aid of the hypnosis and the second according to a standard AS (SAs) protocols. The clinical, radiological and surgical data of two comparable procedures cohorts were retrospectively examined for the present study. All surgeries in Group A were performed with a HAs protocol. Procedures belonging to Group B were performed with a SAs protocol. Endpoints: to compare 1. Incidence of complications in the immediate postoperative period, 2. Clinical and neurological status in the immediate postoperative period and 30 days after surgery, 3. Duration of surgical interventions, 4. Extent of Resection (EOR). The final cohort is composed of 15 procedures; 6 belonging to Group A and 9 to Group B. The different methods outline statistically comparable results from the clinical (Neurological outcomes) both in the postoperative period and one month after surgery and from the surgical point of view (comparable EOR). The incidence of complications is comparable either. The duration of the procedures was significantly longer in HAs group. Hypnosis is a promising approach to increasing the duration of intraoperative "testability" of patients at the price of a longer operative time. A specific professional is needed to induce hypnosis in the difficult intraoperative setting.

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1. Introduction

1.1. Background and rationale

AS supported by Intraoperative Neurophysiological monitoring (ION) and Intraoperative Neuropsychological Testing (IoNT) is becoming the worldwide gold-standard for the surgical management of intrinsic lesions involving the eloquent areas [1–6]. Since a wide range of IoNT batteries can be intraoperatively performed [7], thus preserving the Cognitive functions, the undisputed advantage of AS still appears to be its extreme versatility and accuracy.

An Hypnosis aided Awake Surgery (HAs) protocol has been previously described [8] as extremely promising approach to 1. Reduce the Awake Surgery-related psychological trauma, 2. Increase the amount of time, in minutes, in which the patient is suitable for the IoNT, thus postponing the appearance of Fatigue phenomena, 3. Decrease the incidence of intraoperative Pain and Discomfort episodes. Data concerning the effect of HAs on the Extent of Resection (EOR) and the incidence of Dissociative psychiatric phenomena deserve further investigations because burdened by a more significant statistical uncertainty.

The main limitation of the previous investigation, was that the conclusions we produced derived from a vast number of observations coming from an extensive review of the relevant Literature, in other words, the data, presented an unavoidable "Review" bias

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which had, in our opinion a specific weight especially concerning variables such as the EOR, since in the only previously available paper [9], EOR was reported concerning a cohort of exclusive Low Grade Glioma (LGG) patients, and this implies commonly, a completely different approach to the problem of preserving the functions and thus in the strategy to reach the maximal safe feasible resection in comparison to the High Grade Glioma (HGG) patients, also considering the completely different growth pattern and tendency to infiltrate the white matter eloquent bundles between HGG and LGG [10].

1.2. Background and rationale

Purpose of the present study is therefore to overcome the aforementioned bias by means of a non-randomized retrospective comparison between two cohorts of Glioma patients operated on with an HAs (Group A) or with a Standard AS approach (SAs, Group B). The main objective is to analyze the safety-effectiveness profile of HAs as compared to Standard AS in regards to

1 Incidence of perioperative (intraoperative and immediate postoperative) complications,
2 Clinical and Neurological short-term outcomes, and thus its effect on the suitability for further adjuvant treatments
3 Length of the surgical procedures and
4 Its effect on Extent or Resection.

2. Materials and methods

2.1. Study design and setting

We retrospectively reviewed a consecutive series of patients who underwent surgery for resection of primary intracerebral neoplasms in two different in the period ranging between April 2017 and January 2019 for the patients operated on with Has protocol and between April 2018 and January 2019 for the cohort treated with SAs.

All the surgical procedures were performed in Awake Surgery (As): according to the specific As approach, the cases were subsequently divided and respectively assigned to two subgroups, Group A, including 6 procedures performed with the aid of an original HAs (Hypnosis Awake Surgery) protocol, and 9 cases, composing Group B operated on by means of a Standard As (SAs) protocol.

2.2. Participants and eligibility

All the surgical procedures composing the final cohort matched the following general exclusion criteria

1. Independently from the specific preoperative neurological impairment disclosed by any patients, we excluded patients suffering from cognitive impairment that could hinder a complete and unconditioned understanding of the informed consent and of course a full understanding of the IoNT.
2. Multifocal lesion, requiring more than a single craniotomy for resection were excluded from the final cohort
3. Incomplete or wrong data on clinical, radiological, and surgical records and/or lost to follow-up.

2.3. Data sources and variables

Any patient, at admission underwent a general medical, radiologic, neurologic and a psychological evaluation with a preliminary interview, meant just to rule out the presence of major cognitive dysfunctions to attest the feasibility of the awake IoNT and the preservation of the capacity of discernment in order to understand the extensive explanation provided during the informed consent.

We furthermore considered the following variables: sex, age, histology, side and location of the lesion, extent of resection (EOR), surgical and clinical/neurological complications. The neurological functions were finely evaluated and recorded in three different times, at admission (preoperative), in the first postoperative day, and at the one-month outpatient re-evaluation. In case the hospitalization lasted more than 30 days, the clinical records of the 30th days was recorded.

A special focus was centred on the early postoperative (30 days) neurological results, because of their importance in influencing the accessibility to further adjuvant treatments, in order to assess the real postoperative conditions of such patients, with special regards to the *variation* of the neurological conditions before and after surgery. The occurring of a new or worsening of a pre-existing neurologic deficit was defined as "transient" if it completely resolved without sequelae, or without compromising the final functional status, within the 30th days from surgery: such a variable was coded in dichotomous term (0/1 – No/Yes). Conversely a neurologic deficit was judged as "permanent" if did not show a trend towards resolution within the same span of time and subsequently, the clinical evaluation was performed by means of a three-step ordinal scale (and variable) as follows: "0 = Worsened, 1 = Stable, 2 = Improved".

Furthermore, the incidence of surgical or medical complications related to surgery was recorded either. Among the surgical complications we included the intracranial hematoma/venous infarction, the presence of postoperative hydrocephalus, infection of the surgical wound or of the surgical cavity. Conversely, postoperative severe infections as hospital-acquired pneumonia, sepsis, failures related to postoperative medications, were recorded among the medical complications. Time to resolutions of such complications was recorded also.

2.4. Surgical timing

While revising our previous experience, as also immediately underlined by other authors [8,11] it was clear that one of the most salient features when applying the HAs, is the extremely prolonged suitability of the patients to IoNT, thus obtaining an intraoperative accurate brain mapping, with an obviously enhanced tolerability to the awake surgery conditions, reached at a price of prolonging the surgical time. Obviously, there is a portion of the total required time which is exclusively devoted to the induction of the "Hypnosis", and which varies (also widely) from a patient to another. At that time, the clinical and Literature analysis was based on the number of minutes starting from the beginning of the IoNT, to the end of IoNT.

The present analysis is meant to move a step further, by evaluating and confronting the total amount of time (in minutes), spent within the Operative Theatre, as indicative of the real time needed from the beginning to the end of a surgery.

2.5. Extent of resection

Extent of Resection (EOR), was systematically recorded analyzing the early postoperative Brain MRI scans, which the patient underwent in both Institutions, within 48 h after surgery, by means of a manual segmentation realized with a dedicated software. Extent of Resection (EOR) was coded in a three-step ordinal variable as reported elsewhere [12,13]): gross total resection (GTR $\leq$ 2 mm^3 residual lesion); subtotal or near total resection (STR $\geq$ 2 mm^3).

2.6. Preoperative and operative protocol

For each one of the patients belonging to the two different cohorts, a preoperative magnetic resonance imaging (MRI) scans were performed with the aid of a standard high-field 1.5-T magnet to achieve the diagnosis and exact localization of the tumour with its anatomical relationships for surgical planning and Neuronavigation. The following sequences are routinely performed: T2-weighted, FLAIR, isotropic volumetric T1-weighted magnetization prepared rapid gradient-echo (MPRAGE) with Gadolinium, diffusion weighted imaging (DWI) sequences, and functional magnetic resonance imaging (fMRI) complete the protocol to depict the location of the language and motor functions, in both Institutions.

All surgical procedures of both subgroups were performed in equipped with an infrared-based Neuronavigator and Operative Microscope.

2.7. Hypnosis awake surgery

The hypnosis technique used is described in detail elsewhere [8]. The hypnosis technique used in this protocol is based on the studies performed by Milton Erickson. The objective of this technique is to create a transient state of dissociation in the patient, in order to preserve the possibility to complete the tasks proposed during the IoNT while causing amnesia of the entire surgical procedure. More specifically, the hypnotherapist leads the patient to a "safe place", in other words through the hypnotic trance the patient is convinced that he/she is not undergoing a surgical operation, but he/she is living a comfortable situation, such as being at home with their relatives. The attention of the patient is focused on creating the appropriate mind setting in order to receive the hypnotic suggestions by means of verbal and physical communication between the hypnotherapist and the patient. A particular feature of this technique is that hypnotic suggestions are not imposed to the patient by the therapist, on the contrary the "safe place" is chosen in concert between the patient and the hypnotherapist and all the suggestions are not lived by the patient as an imposition. After the induction of the hypnotic trance, a careful regional anesthesia with total nerve block of the scalp, of the surgical incision and of course of the Mayfield clamp points preceded the skull clamp positioning. The drugs used for the regional anesthesia were Mepivacaine 2% and Ropivacaine 10 mg/ml. A minimally invasive craniotomy drawn with the aid of the Neuronavigator was subsequently performed and, after dural opening without performing local anesthesia, a microsurgical resection of the lesion was matched with a cortical mapping of the language and motor function. Both Motor and Language functions were mapped through a negative mapping approach in standard fashion [3–6]; language function evaluation relied on the Naming, Comprehension and Repetition subscales of the BADA test [14]. Sedative drugs such as Midazolam or Propofol were not administered during any procedure.

2.8. Standard (Full) awake surgery

On the first step a preliminary sedation was obtained by means of an intravenous infusion. More specifically, an intravenous infusion of Remifentanil 0.003 μgr/kg/min was performed for preventing the eventual insurgence of pain. Careful regional anaesthesia with total nerve block of the scalp, of the surgical incision and of course of the Mayfield clamp points preceded the skull clamp positioning. The drugs used for the regional anesthesia were Mepivacaine 2% and Ropivacaine 10 mg/ml. The craniotomy was drawn with the aid of the Neuronavigator and, after dural opening, without performing local anesthesia a microsurgical resection of the lesion was matched with a cortical mapping of the language and motor function. Both motor and language functions were mapped through Motor and Somatosensory Evoked Potential and Phase reversal with a cortical electrode strip approach plus a negative mapping approach with mono- and bipolar probes and stimulations ranging between 3 mA and 7 mA; language function evaluation relied on the Naming, Comprehension and Repetition subscales of BADA test [14].

It was judged necessary to stop tumour excision when:

1. The response to the intraoperative neuropsychological testing matched with data retrieved from neuromonitoring outlined a risk for the insurgence of neurological postoperative morbidity,
2. White matter appeared free of disease in any aspect of the surgical cavity and the Neuronavigator disclosed compatible findings,

In conclusion, the decision to stop tumour excision has been taken by giving priority to intraoperative functional findings rather than to anatomical/neuronavigation ones in order to preserve patients' neurological status, even if this choice entailed the presence of residual disease.

2.9. Study size and power of the study

Because of his "treatments comparison" nature, the study size is given by the different approaches performed. As previously stated, we addressed no missing data, rather we excluded a potentially incomplete variable because of its supposed critical impact on the reliability of the final statistical analysis. A source of bias is expected from the size of the sample, because of its limited number of records. Post hoc statistical estimated power is $1\text{-}\beta = 0.74$ (for $\alpha = 0.2$; effect size, 0.8), in other words, with relative reliability, the clearest effects can be disclosed, stronger effects emerge from the analyses.

The sample was analysed with SPSS v18 (SPSS Inc., released 2009., PASW Statistics for Windows, Version 18.0, Chicago, Illinois, USA) to outline potential correlations between the investigated variables. The comparison between outcome variables was made with paired samples t-tests and One-Way ANOVA tests. Concerning the comparison between the results of HAs and SAs protocol we used the Chi^2 test. Threshold of statistical significance was considered $p < 0.05$.

2.10. Ethical issues

Institutional Review Board of our Institutions expressed favorable opinion towards the surgical informed consent. All the patients included in the final cohort gave informed written explicit consent to surgical procedure after appropriate information. The protocol of the study did not include deviations from the worldwide recognized gold standard in diagnosis and treatment of these patients. Clinical and radiological data reported are completely anonymized. Our work is coherent with the ethical standards proposed in the Helsinki declaration of Human Rights.

3. Results

A total of 15 surgical procedures for the treatment of primary brain neoplasms, have been performed in our Neurosurgical Departments: 6 belonged to Group A (HAs) and 9 to Group B (SAs), in a period ranging from April 2017 to January 2019 for Group A and from April 2018 to January 2019 for Group B.

A total of 9 patients were male and 6 were female. The average age for HAs and SAs subgroups were respectively 52.1 years (22–

Table 1
Patient's demographics.

Patient (N°, sex, age)	Diagnosis	Side	Localization	Surgical time (min)	Pre-operative deficit	Intraoperative complications	Post-operative deficit	Post-operative deficit at 1 month
1, M, 41	Glioblastoma WHO 4	Left	Temporal	420	Mixed aphasia (motor prevalent), mouth and facial apraxia, impossibility to execute complex orders due to comprehension impairment. Slight drifting of the superior right limb during Mingazzini Test	none	none	none
2, F, 70	Glioblastoma WHO 4	Left	Temporal	300	Sensorial aphasia. Retrogade amnesia.	none	none	none
3, F, 72	Glioblastoma WHO 4	Right	temporal	360	Left emiparesis, superior limb > inferior limb . Positivity to Mingazzini test at superior limbs	none	haemotma of the surgical site	resolution of the haematoma
4, M, 41	Glioblastoma WHO 4	Left	temporal	270	Negative	none	none	none
5, M, 22	Astrocytoma WHO II	Right	frontal	360	Negative	none	none	none
6, M, 41	Glioblastoma WHO 4	Left	Temporal	540	Negative	none	none	none
7, F, 57	Oligodendroglioma G II, 1p19q codeleted, Mib 1 8%, MGMT metilation, IDH 1 mutated, IDH 2 wt	Left	fronto temporo insular	300	semantic paraphasia	nessuna	aphasia	resolution of aphasia
8, M, 71	GBM GIV, Mib 1 25%, 1p not deleted, 19q deleted, MGMT not metilatted, IDH 1/2 wt	Left	temporal	240	afasia nominum, amnesia	none	none	resolution of aphasia
9, F, 56	Diffuse astrocytomaGII, Mib 1 3%, MGMT metilation, IDH 1 mutated, IDH 2 wt, 1p notdeletedo, 19q deleted	Left	temporal	220	afasia nominum	none	none	none
10, F, 60	GBM GIV, Mib 1 40%, IDH wt, MGMT not mutated	Left	fronto temporo thalamic insular	260	afasia nominum, amnesia	none	hemiplegia, aphasia	persistance of phasic and right leg motor deficit
11, M, 58	GBM GIV, Mib 1 30%, MGMT metilation, IDH 1/2 wt, high amplification of EGFR	Left	temporal	180	aphasia	none	worsening of aphasia	improvement of aphasia
12, M, 34	Diffuse Astrocytoma GII, Mib 1 4%, MGMT metilatiom, IDH 1 mutato, IDH 2 wt,1p not deleted, 19q deleted	Left	frontal	210	none	epilepsy	articular apraxia, aboulia, aphasia	improvement of all post-operative deficits
13, F, 39	GBM GIV, Mib 1 70%, EGFR amplificated, IDH wt	Left	temporal	180	none	none	none	none
14, M, 71	GBM G IV, Mib 1 20%, MGMT metilated, IDH 1/2 wt	Left	frontal	200	cognitive and motor slowing	none	slight aphasia	none
15, M, 43	Oligodendroglioma G II, Mib 1 8%, 1p19q codeleted, Mib, IDH 1 mutated, IDH 2 wt	Right	parieto occipital	220	lipotimia, apraxia	none	none	none

GBM: Glioblastoma Multiforme; **wt:** wild type; **HAs:** Hypnosis-aided Awake Surgery; **SAs:** Standard Awake Surgery; **WHO:** World Health Organization.

72) and 54.3 years (34–71) with no statistically significant difference between groups (p = .451).

The most common location was the temporal lobe (9/15 patients) followed by frontal lobe (3/15 patients) localization. In two cases belonging to Group B there was respectively a thalamic and extensive insular involvement. A total of 12 patients (80%) presented left lesions while in 3 cases it was located in the right hemisphere. Temporal eloquent areas were peculiarly involved in respectively 4/6 patients belonging to Group A and 4/9 patients belonging to Group, with no statistically significant differences (p = .378). Ten out of fifteen patients were suffering from Glioblastoma Multiforme (66.7%), 3 and 2 patients were suffering from Diffuse Astrocytoma grade II and Oligodendroglioma grade II. The incidence of HGG glioma did not show a statistically different incidence (p = .294).

From a neurological perspective a total of 7/15 patients were suffering from language impairments among which: 3 obvious Aphasias, 3 Naming problems (Aphasia Nominum), and 1 Semantic Paraphasia, all of these disturbances prevented a functional communication. Two patients presented an Hemiparesis. A total of 5 patients were completely intact. Importantly, the incidence of the language deficits did not significantly differ between the two subgroups (p = .378). All the relevant Details are summarized in Table 1.

1. Incidence of perioperative (intraoperative and immediate postoperative) complications
   We recorded a total of 1 surgical complication in HAs group (1/6 patients, Hematoma of the surgical cavity), determining a worsening of the preoperative neurological conditions, as discussed below). In the SAs group we recorded the incidence intraoperative seizures (1/9 patients). Such feature disclosed no statistical significance (p = .378).
2. Clinical and neurological short-term outcomes
   As disclosed in Table 2, in the early postoperative, we recorded 4 worsening of the neurological conditions, 1 belonging to Group A and 3 and belonging to Group B. A total of 11 patients were stable or improved in respect to the preoperative period. At 30 days, 2 patients had a full recovery thus experiencing a transient deficit; the remaining 2 patients (1 per each subgroup) experience a more stable worsening, pointing at a substantial stability/improvement of the general conditions at 30 days in 86.7% of cases (13/15 patients). The relevant analyses ruled out statistically significant differences, both for what concerns the immediate postoperative period and results at 30 days (p = .506 and p = .487 respectively; Figs. 1–3).
3. Extent of resection
   SAs and HAs disclosed no statistically significant difference in terms of EOR for what the present cohort is concerned, their effectiveness in promoting a maximal safe resection was comparable. Indeed, we recorded two cases of STR, one per each subgroup (p = .657; Fig. 4).
4. Length of the surgical procedures
   The (all inclusive) length of the surgical procedures of patients belonging to HAs group, was, as previously discussed [8,11], long; from a statistical point of view the difference is outstandingly clear (p = .0001, Fig. 5). The difference attained 202 min averagely (223.3 ± 38.7 versus 425.8 ± 105.5). The reasons justifying this peculiar feature deserve an extensive mention in Discussion paragraph.

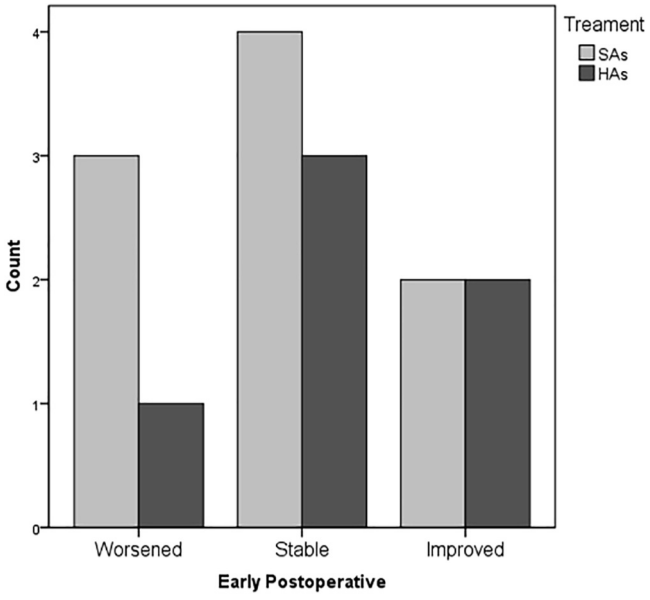


Fig. 1. Bar Chart disclosing the Early Postoperative Neurological Outcomes.

Table 2
Neurological Outcomes in the Early Postoperative period and at 30 days.

Early Postoperative				
		Treatment		Total
		SAs	HAs	
Immediate	Worsened	3	1	4
	Stable	4	3	7
	Improved	2	2	4
Total	9	6	15	
One month after surgery				
		Treatment		Total
		SAs	HAs	
One Month	Worsened	1	1	2
	Stable	3	3	6
	Improved	5	2	7
Total	9	6	15	

SAs: Standard Awake surgery; **HAs:** Hypnosis-aided Awake surgery.

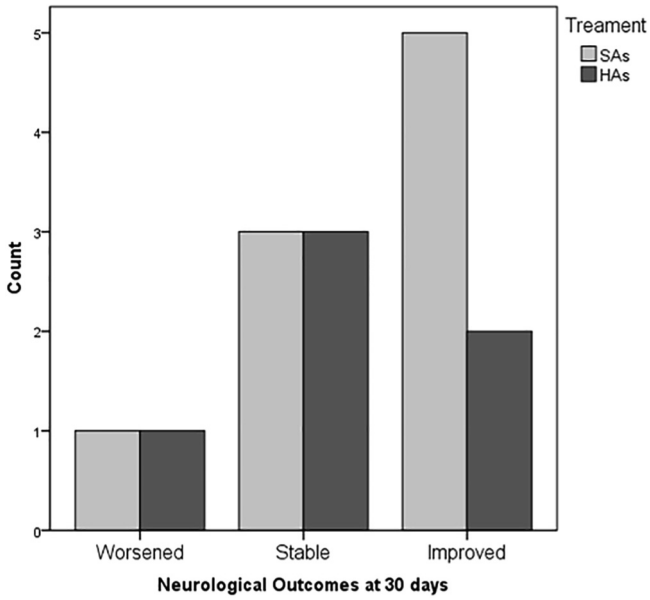


Fig. 2. Bar Chart disclosing the 30th day Neurological Outcomes.

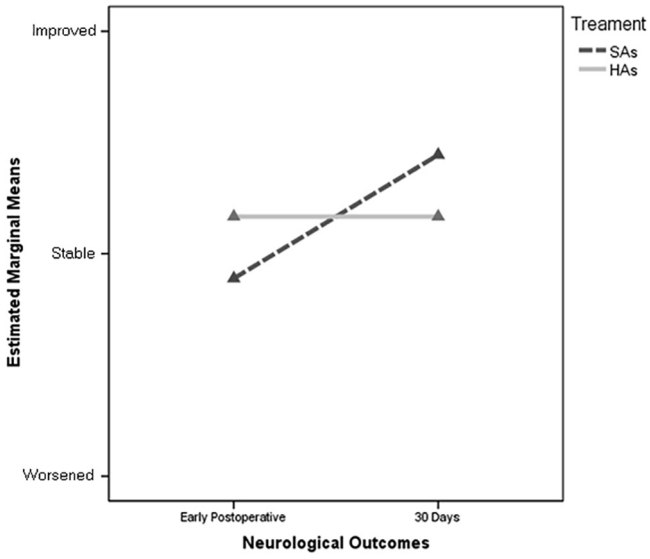


Fig. 3. ANOVA Repeated Measures outlines the substantial superimposability of the clinical outcomes of the two approaches.

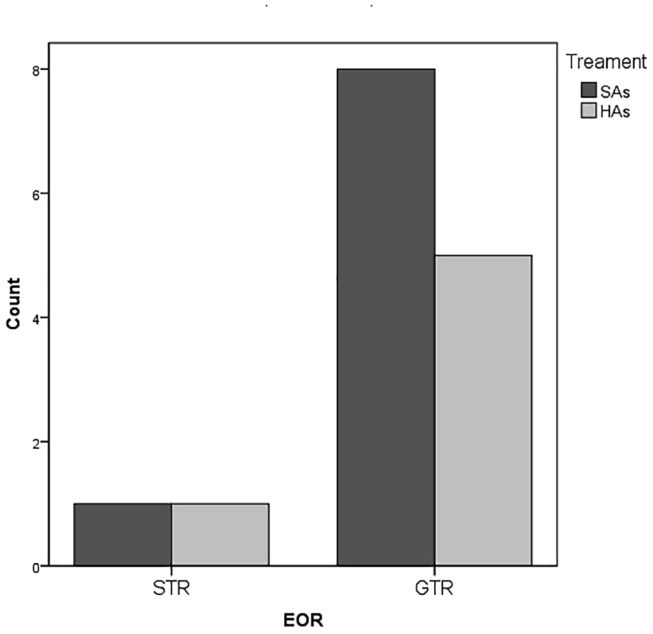


Fig. 4. Bar Chart disclosing comparable outcomes in terms of EOR.

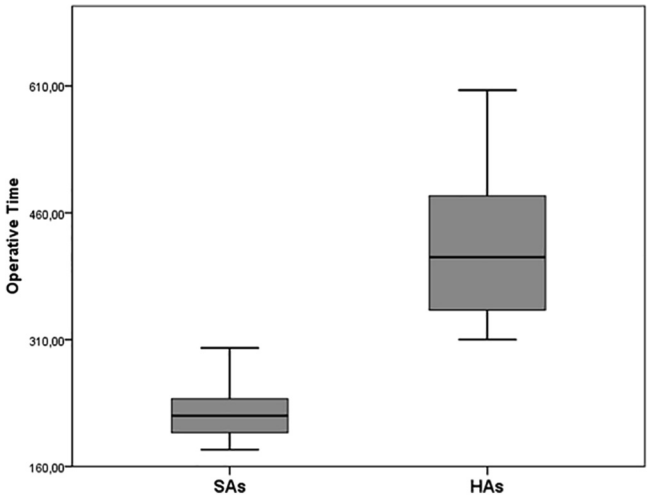


Fig. 5. One Way ANOVA analysis disclosing the statistically significant difference in regards to the length of surgeries.

4. Discussion

Clinical research in this branch of neurosurgery is in continuous development in order to enhance not only the standards of treatment by obtaining better results in terms of survival and neurological status after surgery, but also by improving the psychological effects of such traumatic experience on patients' lives. In fact, the low levels of tolerance towards the IoNT and the visual and auditory perceptions felt by the patients during the awake part of the surgical procedure represent one of the main limitations of SAs [15–17]. The first study to describe an HAs protocol belongs to the group of Zemmoura et al. [9]. In this paper intraoperative discomfort and intraoperative pain were registered during HAs procedures and they showed lower levels of incidence compared to SAs cohorts [9]. Up to date, the protocol described by Zemmoura [9] et al. and the one proposed by Frati et al [8] represent the only studies in Literature that analyse data regarding psychological effects of hypnosis after neurosurgical operations respecting this requisite. The cohort of patients analyzed was obtained by matching the cohort of Zemmoura et al and the one collected for this study and then they were confronted to different SAs cohorts. In this protocol [8], based on the hypnosis model theorized by Milton Erickson [18], the advantages firstly described by Zemmoura et al. of HAs protocols on SAs procedures in terms of patients' satisfaction are confirmed and, furthermore, the intensity of awake surgery-related psychological trauma was studied by means of a universally validated psychological test (Impact of the Event Scale-Revised Version) [19–22]. The possibility to perform just regional anaesthesia without using sedative drugs, the lack of memories related to surgery as consequence of the hypnotic trance and, above all, the possibility to perform the IoNT and brain mapping with an high grade of tolerance from the patients and for a

longer time span represent the major advantages that emerged from the preliminary analysis of this HAs protocol [8]. On the other hand, the results of the study on the intensity of awake surgery-related psychological trauma showed that the incidence of post-traumatic experience was of 18.36% and the 36.74% of the patients experienced at least a transient dissociation state after surgery. These findings suggest that there is a concrete risk of psychological consequences related to HAs protocols.

Regarding safety and effectiveness of HAs protocols in terms of surgical, clinical and neurological outcomes it is possible to affirm that HAs and SAs protocols guarantee the same results (Figs. 1–3). On the base of the results of the present study is clear that the use of hypnosis does not modify the course of surgery, since it is possible to use the same technologies and surgical approaches used in SAs procedures. However, the relevant finding regards the difference in length of surgical procedures between Group A and B (Fig. 4). As observed by Pardey Bracho [11] et al., IoNT time span and, consequentially, length of surgical procedures in the HAs protocol [8] is significantly longer than SAs procedures. The results from the present study outlines this difference and the final longer length for HAs procedures represents the *summa* of different causes. First of all, the induction of hypnotic trance takes averagely 50 min to the hypnotherapist in order to reach the transient dissociation state that is the essential base of the hypnosis model proposed by Erikson [19]. Moreover, the application of this HAs protocol is a novelty [8], therefore the long duration of the IoNT is mostly explained by the mandatory necessity to be very careful with the use of the technique [23]. More specifically, we performed an extremely accurate IoNT by repeating the tests multiple time and by investigating more cognitive functions and modules than those strictly related to the areas involved by the lesions. At the same time, this reiterative method was important in order to retrieve more data regarding the safety-tolerability profile of the technique for our patients. The final findings showed that hypnosis allows the patient to tolerate the IoNT for longer time, ensuring the quality of responses from the tests that is obviously required in order to perform the safest and more effective resection of the lesion. Nevertheless, it is clear that a complete analysis of all the factors related both to patients and to the hypnotic procedure that influence psychological outcome needs to be conducted through further studies.

5. Limitations and advantages of the present study

The main limitations of the present study rely in the exiguity of the sample, which nevertheless was suitable to detect obvious statistical differences between the subgroups, and its retrospective nature. The purpose of the present study was to overcome the "Literature-bias" of the previous investigation, by collecting and comparing data in a direct and retrospective fashion, with extremely focused analyses concerning precise outcome variables. This is conversely the main advantage of the present paper: focused comparison to enhance the statistical power of the study thus improving the quality of the conclusions.

6. Conclusions

The comparison between HAs and SAs demonstrated no statistically significant difference regarding the main safety-effectiveness outcome variables commonly considered as critical in Glioma Surgery: EOR, Incidence of Complications and Clinical and Neurological outcomes. The only parameter showing an obvious statistically significant difference was the Length of the surgical procedure. This data outlines a longer duration of the surgeries related to a longer Hypnosis induction required time and reflects in a longer suitability of the patients for IoNT. Apart the problem of time, one of the main limitations of HAs is that a trained professional with solid Hypnosis-inductive skills is mandatory to perform this approach.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript. The authors confirm their adherence to ethical standards and have NO financial disclosures that would be a potential conflict of interest with this publication.

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